Solutions to Axler's *Linear Algebra Done Right*, 4th Edition

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Contents

Preface **3**

1 Vector Spaces **4**
1A $\mathbf{R}^n$ and $\mathbf{C}^n$ 4
1B Definition of Vector Space 9
1C Subspaces 14

2 Finite-Dimensional Vector Spaces **24**
2A Span and Linear Independence 24
2B Bases 33
2C Dimension 40

3 Linear Maps **51**
3A Vector Space of Linear Maps 51
3B Null Spaces and Ranges 62
3C Matrices 78
3D Invertibility and Isomorphisms 88
3E Products and Quotients of Vector Spaces 99
3F Duality 108

4 Polynomials **124**

5 Eigenvalues and Eigenvectors **130**
5A Invariant Subspaces 130
5B The Minimal Polynomial 146
5C Upper-Triangular Matrices 162
5D Diagonalizable Operators 169
5E Commuting Operators 182

6 Inner Product Spaces **187**
6A Inner Products and Norms 187
6B Orthonormal Bases 205
6C Orthogonal Complements and Minimization Problems 220

7 Operators on Inner Product Spaces **231**
7A Self-Adjoint and Normal Operators 231
7B Spectral Theorem 246
7C Positive Operators 258
7D Isometries, Unitary Operators, and Matrix Factorization 268
7E Singular Value Decomposition 278
7F Consequences of Singular Value Decomposition 287

8 Operators on Complex Vector Spaces **303**
8A Generalized Eigenvectors and Nilpotent Operators 303
8B Generalized Eigenspace Decomposition 313
8C Consequences of Generalized Eigenspace Decomposition 324
8D Trace: A Connection Between Matrices and Operators 330

A Appendix: Alternate Proofs of Exercises **333**
A1 Vector Spaces 333
A2 Finite-Dimensional Vector Spaces 334
A3 Linear Maps 335
A4 Polynomials 340
A5 Eigenvalues and Eigenvectors 341
A6 Inner Product Spaces 344
A7 Operators on Inner Product Spaces 346
A8 Generalized Eigenvectors and Nilpotent Operators 350

8C Consequences of Generalized Eigenspace Decomposition

Manual Remark. *The terms* superdiagonal, second superdiagonal, *and* subdiagonal *refer to the entries immediately above, two above, and immediately below the main diagonal of a matrix, respectively. For* 1*-by-*1 *blocks, all three diagonals are empty, while for* 2*-by-*2 *blocks, the second superdiagonal is empty, so claims involving the entries of these empty diagonals hold vacuously.*

Exercise 8C1

Suppose $T \in \mathcal{L}(\mathbf{C}^3)$ is the operator defined by $T(z_1, z_2, z_3) = (z_2, z_3, 0)$. Prove that T does not have a square root.

Solution. Note that $T^2(z_1, z_2, z_3) = (z_3, 0, 0)$ and $T^3 = 0$. Suppose $S \in \mathcal{L}(\mathbf{C}^3)$ is such that $S^2 = T$. Then $S^6 = (S^2)^3 = T^3 = 0$, so S is nilpotent. By 8.16, $S^3 = 0$, but $T^2 = S^4 = SS^3 = 0$, which contradicts $T^2 \neq 0$. Therefore, T does not have a square root. ∎

Exercise 8C2

Define $T \in \mathcal{L}(\mathbf{F}^5)$ by $T(x_1, x_2, x_3, x_4, x_5) = (2x_2, 3x_3, -x_4, 4x_5, 0)$.

(a) Show that T is nilpotent.

(b) Find a square root of $I + T$.

Solution.

(a) With respect to the standard basis of $\mathbf{F}^5$, the matrix $\mathcal{M}(T)$ of T on and below the diagonal is 0, so T is nilpotent by 8.18.

(b) By part (a) and 8.16, $T^5 = 0$. As in the proof of 8.39, we may construct the polynomial

$$p(z) = 1 + \frac{1}{2}z - \frac{1}{8}z^2 + \frac{1}{16}z^3 - \frac{5}{128}z^4.$$

Squaring $p(z)$ gives

$$p^2(z) = 1 + z + z^5 q(z)$$

for some $q \in \mathcal{P}(\mathbf{F})$. Let $S = p(T)$, so $S^2 = I + T + T^5 q(T) = I + T$ by 5.17(a) and $T^5 = 0$. ∎

Exercise 8C3

Suppose V is a complex vector space. Prove that every invertible operator on V has a cube root.

Manual Remark. *The coefficients in the recursion below are given by* $a_j = \binom{1/3}{j}$ *for* $j \geq 1$.

Solution. Let $T \in \mathcal{L}(V)$ be invertible with distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ and let $n = \dim V$. By 5.7, $\lambda_k \neq 0$ for each $k = 1, \ldots, m$. Let $G_k = G(\lambda_k, T)$ for each $k = 1, \ldots, m$. Then the generalized eigenspace decomposition of V is

$$V = \bigoplus_{k=1}^{m} G_k.$$

By 8.22(a), we may define $N_k = (T - \lambda_k I)|_{G_k} \in \mathcal{L}(G_k)$ for each $k = 1, \ldots, m$. Since N_k is nilpotent on G_k by 8.22(b), and $\dim G_k \leq n$, we have $N_k^n = 0$ by 8.16. As in the proof of 8.39 with a cubic in place of a square, set $a_1 = \frac{1}{3}$. For each $2 \leq j \leq n-1$, the coefficient of z^j in the expansion $(1 + a_1 z + \cdots + a_j z^j)^3$ is $3a_j + f_j(a_1, \ldots, a_{j-1})$ for some polynomial f_j in $a_1, \ldots, a_{j-1}$. Choosing a_j such that $3a_j + f_j(a_1, \ldots, a_{j-1}) = 0$, we can recursively determine $a_2, \ldots, a_{n-1}$. We then obtain the polynomial

$$p(z) = 1 + a_1 z + \cdots + a_{n-1} z^{n-1}$$

such that $p^3(z) = 1 + z + z^n r(z)$ for some $r \in \mathcal{P}(\mathbf{F})$. By the fundamental theorem of algebra, the polynomial $z^3 - \lambda_k$ has a zero in $\mathbf{C}$ for each $k = 1, \ldots, m$, say μ_k. Define the operators $S_k = \mu_k p(\lambda_k^{-1} N_k)$ on G_k for each k. By 5.17(a) and $N_k^n = 0$, it follows that $S_k^3 = (\mu_k p(\lambda_k^{-1} N_k))^3 = \lambda_k I + N_k = T|_{G_k}$. For each $v \in V$, by the generalized eigenspace decomposition, we may uniquely write $v = v_1 + \cdots + v_m$ with $v_k \in G_k$ for each $k = 1, \ldots, m$. Define $S \in \mathcal{L}(V)$ by $Sv = S_1 v_1 + \cdots + S_m v_m$. Since $Sv_k = S_k v_k \in G_k$ for each k, we have $S^3 v_k = S_k^3 v_k = T|_{G_k} v_k = T v_k$ for each k, and thus $S^3 = T$ on all of V. ∎

Exercise 8C4

Suppose V is a real vector space. Prove that the operator $-I$ on V has a square root if and only if $\dim V$ is an even number.

Solution.
$\boxed{\Rightarrow}$ Let $S \in \mathcal{L}(V)$ be such that $S^2 = -I$. Then $\operatorname{null}(S^2 + 0S + I) = \operatorname{null}(-I + I) = \operatorname{null}(0) = V$. Since $0^2 < 4$, 5.33 gives that $\dim V$ is even.
$\boxed{\Leftarrow}$ Let $\dim V = 2n$ for some positive integer n, and let V have a basis $v_1, \ldots, v_{2n}$. By the linear map lemma, there exists a linear map $S \in \mathcal{L}(V)$ such that $Sv_{2k-1} = v_{2k}$ and $Sv_{2k} = -v_{2k-1}$ for each $k = 1, \ldots, n$. Then $S^2 v_{2k-1} = Sv_{2k} = -v_{2k-1}$ and $S^2 v_{2k} = S(-v_{2k-1}) = -v_{2k}$ for each $k = 1, \ldots, n$. Since S^2 and $-I$ agree on a basis of V, the uniqueness part of the linear map lemma implies that $S^2 = -I$. ∎

Exercise 8C5

Suppose $T \in \mathcal{L}(\mathbf{C}^2)$ is the operator defined by $T(w, z) = (-w - z, 9w + 5z)$. Find a Jordan basis for T.

Solution. Let $N = T - 2I$. Then $N(w, z) = (-3w - z, 9w + 3z) = (-3w - z)(1, -3)$, so

$$N^2(w, z) = (-3w - z)N(1, -3) = 0.$$

Let $v = (0, -1)$. Then $Nv = (1, -3) \neq 0$, so by Exercise 8A2, v, Nv is a linearly independent list of length 2, and thus is a basis of $\mathbf{C}^2$ by 2.38. We have $TNv = 2Nv$ and $Tv = Nv + 2v$, so

$$\mathcal{M}(T, (Nv, v)) = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

Therefore, the list $(1, -3), (0, -1)$ is a Jordan basis for T. ∎

Exercise 8C6

Find a basis of $\mathcal{P}_4(\mathbf{R})$ that is a Jordan basis for the differentiation operator D on $\mathcal{P}_4(\mathbf{R})$ defined by $Dp = p'$.

Solution. Define $v = \frac{x^4}{4!}$, so $D^4 v = 1$ and $D^5 v = 0$. By Exercise 8A2, the list $v, Dv, D^2 v, D^3 v, D^4 v$ is a linearly independent list of length 5 and thus forms a basis of $\mathcal{P}_4(\mathbf{R})$ by 2.38. Then

$$\mathcal{M}(D, (D^4 v, D^3 v, D^2 v, Dv, v)) = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix},$$

so the list $1, x, \frac{x^2}{2!}, \frac{x^3}{3!}, \frac{x^4}{4!}$ is a Jordan basis for D. ■

Exercise 8C7

Suppose $T \in \mathcal{L}(V)$ is nilpotent and $v_1, \ldots, v_n$ is a Jordan basis for T. Prove that the minimal polynomial of T is z^{m+1}, where m is the length of the longest consecutive string of 1's that appears on the line directly above the diagonal in the matrix of T with respect to $v_1, \ldots, v_n$.

Solution. Let $A_1, \ldots, A_p$ be the blocks as in 8.44, where each A_k is d_k-by-d_k. Let $v_{k,1}, \ldots, v_{k,d_k}$ be the basis vectors of the block A_k for each $k = 1, \ldots, p$. By 8.44,

$$Tv_{k,j} = \begin{cases} \lambda_k v_{k,1} & \text{if } j = 1, \\ \lambda_k v_{k,j} + v_{k,j-1} & \text{if } j > 1. \end{cases}$$

Note that $Tv_{k,1} = \lambda_k v_{k,1}$ with $v_{k,1} \neq 0$. Since T is nilpotent, 8.17(a) implies that $\lambda_k = 0$ for all k. Then $Tv_{k,1} = 0$ and $Tv_{k,j} = v_{k,j-1}$ for $j > 1$, so

$$T^{j-1} v_{k,j} = v_{k,1}, \quad T^j v_{k,j} = 0.$$

The entries above the diagonal between the blocks A_k and A_{k+1} are all 0, so the consecutive string of 1's cannot extend beyond a single block. Therefore, each block A_k contributes a consecutive string of 1's of length $d_k - 1$, so $m + 1 = \max_k d_k$. By 8.18, the minimal polynomial of T is z^r for some r. We now claim that $r = m + 1$.

$\boxed{\leq}$ Since each $v_{k,j}$ has $j \leq d_k \leq m + 1$, we have $T^{m+1} v_{k,j} = T^{m+1-j}(T^j v_{k,j}) = 0$. Therefore, $T^{m+1} = 0$, and 5.29 yields $r \leq m + 1$.

$\boxed{\geq}$ Consider some k such that $d_k = m + 1$. Then $T^m v_{k,m+1} = v_{k,1} \neq 0$, so $T^m \neq 0$. Therefore, $r \geq m + 1$ by 5.29. Combining this with the previous inequality gives $r = m + 1$. ■

Exercise 8C8

Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_n$ is a basis of V that is a Jordan basis for T. Describe the matrix of T^2 with respect to this basis.

Solution. Let $A_1, \dots, A_p$ be the blocks of $\mathcal{M}(T)$ with respect to the basis $v_1, \dots, v_n$ and let λ_k be the eigenvalue corresponding to the block A_k. By 3.43 and Exercise 8B22, $\mathcal{M}(T^2) = \mathcal{M}(T)^2$ is block diagonal with blocks $A_1^2, \dots, A_p^2$. Write $A_k = \lambda_k I + N_k$, where N_k is the matrix with 1's on the superdiagonal and 0's elsewhere. Then $A_k^2 = \lambda_k^2 I + 2\lambda_k N_k + N_k^2$, and N_k^2 has 1's on the second superdiagonal and 0's elsewhere by 3.46. Therefore, each block A_k^2 has λ_k^2 on the diagonal, $2\lambda_k$ on the superdiagonal, 1's on the second superdiagonal, and 0's elsewhere. ■

Exercise 8C9

Suppose $T \in \mathcal{L}(V)$ is nilpotent. Explain why there exist $v_1, \dots, v_n \in V$ and nonnegative integers $m_1, \dots, m_n$ such that (a) and (b) below both hold.

(a) $T^{m_1} v_1, \dots, T v_1, v_1, \dots, T^{m_n} v_n, \dots, T v_n, v_n$ is a basis of V.

(b) $T^{m_1+1} v_1 = \cdots = T^{m_n+1} v_n = 0$.

Solution. By 8.45, nilpotence of T gives a Jordan basis $u_1, \dots, u_{\dim V}$ for T. Let $A_1, \dots, A_n$ be the blocks of $\mathcal{M}(T, (u_1, \dots, u_{\dim V}))$ with diagonal entries $\lambda_1, \dots, \lambda_n$. Let $u_{k,1}, \dots, u_{k,d_k}$ be the basis vectors corresponding to the d_k-by-d_k block A_k. By 5.41 and 8.17(a), $\lambda_k = 0$, so $T u_{k,1} = 0$ and $T u_{k,j} = u_{k,j-1}$ for $2 \le j \le d_k$ by 8.44. Let $v_k = u_{k,d_k}$ and $m_k = d_k - 1$ for $1 \le k \le n$.

(a) is satisfied because $T^j v_k = u_{k,d_k - j}$ for $0 \le j \le m_k$, so the vectors $T^{m_k} v_k, \dots, T v_k, v_k$ are exactly $u_{k,1}, \dots, u_{k,d_k}$ in order.

(b) is satisfied because $T^{m_k+1} v_k = T^{d_k} u_{k,d_k} = 0$ for each k, as $u_{k,1} \in \operatorname{null} T$. ■

Exercise 8C10

Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of V that is a Jordan basis for T. Describe the matrix of T with respect to the basis $v_n, \dots, v_1$ obtained by reversing the order of the v's.

Solution. Let $A_1, \dots, A_p$ be the Jordan blocks of $\mathcal{M}(T, (v_1, \dots, v_n))$ with diagonal entries $\lambda_1, \dots, \lambda_p$. Let $v_{k,1}, \dots, v_{k,d_k}$ be the basis vectors among $v_1, \dots, v_n$ corresponding to A_k in order. By 8.44, $T v_{k,1} = \lambda_k v_{k,1}$ and $T v_{k,j} = v_{k,j-1} + \lambda_k v_{k,j}$ for $2 \le j \le d_k$. In the basis $v_n, \dots, v_1$, the basis vectors corresponding to A_k appear in the reverse order $v_{k,d_k}, \dots, v_{k,1}$, with the blocks appearing in the order $A_p, \dots, A_1$. Writing $w_\ell = v_{k,d_k+1-\ell}$ for $1 \le \ell \le d_k$, we have $T w_\ell = \lambda_k w_\ell + w_{\ell+1}$ for $1 \le \ell \le d_k - 1$ and $T w_{d_k} = \lambda_k w_{d_k}$. Therefore, $\mathcal{M}(T, (v_n, \dots, v_1))$ is block diagonal with the blocks $A_p^{\mathrm{t}}, \dots, A_1^{\mathrm{t}}$. Each block A_k^{t} has λ_k on its diagonal, 1's on its subdiagonal, and 0's elsewhere. ■

Exercise 8C11

Suppose $T \in \mathcal{L}(V)$. Explain why every vector in each Jordan basis for T is a generalized eigenvector of T.

Solution. Let $v_1, \dots, v_n$ be a Jordan basis for T. Let $A_1, \dots, A_p$ be the blocks of $\mathcal{M}(T, (v_1, \dots, v_n))$ with diagonal entries $\lambda_1, \dots, \lambda_p$. Let $v_{k,1}, \dots, v_{k,d_k}$ be the basis vectors corresponding to A_k in order. By 8.44, $(T - \lambda_k I) v_{k,1} = 0$ and $(T - \lambda_k I) v_{k,j} = v_{k,j-1}$ for $2 \le j \le d_k$. Hence, λ_k is an eigenvalue of T and $(T - \lambda_k I)^j v_{k,j} = 0$ for $1 \le j \le d_k$. Since $v_{k,j} \ne 0$, it follows that $v_{k,j}$ is a generalized eigenvector of T corresponding to λ_k. ■

Exercise 8C12

Suppose $T \in \mathcal{L}(V)$ is diagonalizable. Show that $\mathcal{M}(T)$ is a diagonal matrix with respect to every Jordan basis for T.

Solution. Let $v_1, \dots, v_n$ be a Jordan basis for T. Let $A_1, \dots, A_p$ be the blocks of $\mathcal{M}(T, (v_1, \dots, v_n))$ with diagonal entries $\lambda_1, \dots, \lambda_p$. Let $v_{k,1}, \dots, v_{k,d_k}$ be the basis vectors corresponding to the block A_k, and define $U_k = \operatorname{span}(v_{k,1}, \dots, v_{k,d_k})$. By 8.44, T maps the vectors $v_{k,1}, \dots, v_{k,d_k}$ into U_k, so U_k is invariant under T, and $\mathcal{M}(T|_{U_k}, (v_{k,1}, \dots, v_{k,d_k})) = A_k$. By 5.65, $T|_{U_k}$ is diagonalizable for each k. Since A_k is upper triangular with λ_k on the diagonal, 5.41 yields that λ_k is the only eigenvalue of $T|_{U_k}$. By 5.55, $U_k = E(\lambda_k, T|_{U_k})$, so $T|_{U_k} = \lambda_k I$, and hence $A_k = \lambda_k I$. Since this holds for each block A_k, it follows that $\mathcal{M}(T, (v_1, \dots, v_n))$ is diagonal. ■

Exercise 8C13

Suppose $T \in \mathcal{L}(V)$ is nilpotent. Prove that if $v_1, \dots, v_n$ are vectors in V and $m_1, \dots, m_n$ are nonnegative integers such that

$$T^{m_1}v_1, \dots, Tv_1, v_1, \dots, T^{m_n}v_n, \dots, Tv_n, v_n \text{ is a basis of } V$$

and

$$T^{m_1+1}v_1 = \cdots = T^{m_n+1}v_n = 0,$$

then $T^{m_1}v_1, \dots, T^{m_n}v_n$ is a basis of $\operatorname{null} T$.

Axler Note. *This exercise shows that $n = \dim \operatorname{null} T$. Thus, the positive integer n that appears above depends only on T and not on the specific Jordan basis chosen for T.*

Solution. Since $T^{m_k+1}v_k = 0$ for each k, $T^{m_k}v_k \in \operatorname{null} T$ for each k. Observe that $T(T^j v_k) = T^{j+1}v_k$ for each k and $0 \le j \le m_k$, where $T^{m_k+1}v_k = 0$. Therefore, the list $Tv_1, \dots, T^{m_1}v_1, \dots, Tv_n, \dots, T^{m_n}v_n$ spans $\operatorname{range} T$ and is linearly independent since it is part of the given basis. Consequently, the list forms a basis of $\operatorname{range} T$, and hence $\dim \operatorname{range} T = \sum_{k=1}^{n} m_k$. Because $\dim V = \sum_{k=1}^{n}(m_k + 1)$, the fundamental theorem of linear maps gives

$$\dim \operatorname{null} T = \sum_{k=1}^{n}(m_k + 1) - \sum_{k=1}^{n} m_k = n.$$

Since $T^{m_1}v_1, \dots, T^{m_n}v_n$ is part of the given basis, it is a linearly independent list of length n in $\operatorname{null} T$, wherefore it forms a basis of $\operatorname{null} T$ by 2.38. ■

Exercise 8C14

Suppose $\mathbf{F} = \mathbf{C}$ and $T \in \mathcal{L}(V)$. Prove that there does not exist a direct sum decomposition of V into two nonzero subspaces invariant under T if and only if the minimal polynomial of T is of the form $(z - \lambda)^{\dim V}$ for some $\lambda \in \mathbf{C}$.

Solution. Let p be the minimal polynomial of T and let $n = \dim V$.

$\boxed{\Rightarrow}$ Suppose no direct sum decomposition of V into two nonzero subspaces invariant under T exists. By 8.46, there exists a Jordan basis $v_1, \ldots, v_n$ of V with blocks $A_1, \ldots, A_q$. Define U to be the span of the basis vectors corresponding to the first block A_1 and W to be the span of the remaining basis vectors. By 8.44, T maps the basis vectors corresponding to each block into their own span, so U and W are invariant under T. Since $U + W$ contains the basis vectors and $\dim U + \dim W = n$, it follows that $V = U \oplus W$ by 3.94. Since $U \neq \{0\}$, the hypothesis implies that $W = \{0\}$. Hence, $\mathcal{M}(T, (v_1, \ldots, v_n)) = A_1$ is a single n-by-n block. Let $\lambda = \lambda_1$. By 8.44, $(T - \lambda I)v_1 = 0$ and $(T - \lambda I)v_j = v_{j-1}$ for $2 \leq j \leq n$, wherefore $(T - \lambda I)^n = 0$ and $(T - \lambda I)^{n-1} v_n = v_1 \neq 0$. By 5.29, p divides $(z - \lambda)^n$, so $p = (z - \lambda)^r$ for some positive integer $r \leq n$. On the other hand, if $r \leq n - 1$, then $(z - \lambda)^{n-1}$ is a multiple of p, so $(T - \lambda I)^{n-1} = 0$ by 5.29, contradicting $(T - \lambda I)^{n-1} v_n = v_1 \neq 0$. Hence, $r = n$, so $p = (z - \lambda)^n$.

$\boxed{\Leftarrow}$ Suppose $p = (z - \lambda)^n$ for some $\lambda \in \mathbf{C}$. Let $V = U \oplus W$ with U and W invariant under T. Since $(T - \lambda I)^n = 0$ and invariance of U under T gives invariance under $T - \lambda I$, it follows that $(T - \lambda I)^{\dim U} = 0$ on U by 8.16. A similar argument shows that $(T - \lambda I)^{\dim W} = 0$ on W. Let $M = \max(\dim U, \dim W)$. Then $(T - \lambda I)^M = 0$ on $U + W = V$, so $M \geq n$ by 5.29. By 3.94, $\dim U + \dim W = n$, so $M \geq \dim U + \dim W$, forcing $\min(\dim U, \dim W) = 0$. Hence, either $U = \{0\}$ or $W = \{0\}$, proving that no direct sum decomposition into two nonzero invariant subspaces exists. ■